Lesson 6.5 Guided Notes

Session 1 · Writing Algebraic Expressions

I can write an algebraic expression that uses a variable for an unknown amount. · 6.EE.A.2a

Word / Spanish	Meaning — complete the blank	Example
variable / variable	A _____ that stands for an unknown number.	the m in 14m + 35
coefficient / coeficiente	The number _____ by the variable.	In 14m, the coefficient is 14
constant / constante	A number by itself that does not _____.	the 35 in 14m + 35
algebraic expression / expresión algebraica	An expression with at least one _____.	14m + 35

Key idea

An amount that repeats “each” time goes with the _____.

A one-time amount is the _____.

Order matters: “4 less than x” is x _____ 4.

Worked example

A bike shop charges $9 per hour plus a $3 helmet fee. Write an expression for the cost of h hours.

The $9 repeats each hour: 9 × h = _____.

The one-time fee is the constant: _____.

Join the terms with a plus sign.

Answer: The cost is _____ dollars.

You try

1. Write an algebraic expression.
a. 6 more than n _____
b. the product of 4 and k _____

_____

2. Concert tickets cost $10 each plus one $4 fee. Write an expression for the cost of t tickets.

_____

Sum it up: In 9h + 3, the 9 goes with h because ___.

_____

Session 2 · Evaluating Algebraic Expressions

I can evaluate an algebraic expression by substituting a value for the variable. · 6.EE.A.2c

Word / Spanish	Meaning — complete the blank	Example
substitute / sustituir	_____ a variable with a given number.	If b = 7: 6b = 6 × 7
evaluate / evaluar	Find the _____ after you substitute.	6 × 7 − 9 = 33
variable / variable	A letter that stands for a _____.	the b in 6b − 9

Key idea

To evaluate, _____ the value for the variable.

Then use the _____ of operations: powers, then × and ÷, then + and −.

4x means 4 × x. When x = 3, 4x = _____, not 43.

Worked example

Evaluate 7p − 2 when p = 4.

Substitute: 7 × _____ − 2.

Multiply first: 7 × 4 = _____.

Subtract: 28 − 2 = _____.

Answer: 7p − 2 = _____ when p = 4

You try

1. Evaluate when a = 3.
a. 5a + 1 = _____
b. a^2 − 4 = _____

_____

2. A garage charges 3h + 6 dollars for h hours. Find the cost for 5 hours.

_____

Sum it up: When I evaluate 4 + 2p, I multiply first because ___.

_____

Session 1 · Model, reason, and explain

Writing Algebraic Expressions · 6.EE.A.2a

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A rental charges $7 per hour and a one-time $4 fee. Write an expression for h hours. Explain which term repeats and whether this is already an equation.

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: repeats the fixed fee or writes an equation without an equality relationship.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Evaluating Algebraic Expressions · 6.EE.A.2c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Evaluate 6p + 4 when p = 3. Show substitution, evaluate, and connect the terms to a cost story.

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: adds before multiplying or treats 6p as 6+p.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?